

BANYUWANGI, Indonesia

WMO: 969870

Lat: 8.2143S

Lon: 114.3555E

Elev: 16

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	71.9	73.0	64.9	92.3	75.7	66.3	96.8	76.2	12.9	83.4	11.6	82.1	1.4	270	0.308

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	12.1	91.5	78.8	90.6	78.7	89.8	78.7	80.9	88.1	80.3	87.5	79.9	87.1	7.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.8	149.7	84.9	78.2	146.7	84.5	77.9	145.0	84.2	44.4	88.6	43.9	88.1	43.5	87.5	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.2	10.4	9.1	DB	69.5	95.3	1.2	3.8	68.7	98.0	68.0	100.3	67.3	102.4	66.5	105.2
			WB	64.9	83.7	2.2	2.2	63.3	85.3	62.0	86.5	60.8	87.8	59.1	89.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	81.8	81.9	82.3	82.8	82.0	80.5	79.3	79.1	80.4	82.3	83.5	82.6
	DBStd	2.08	1.58	1.55	1.57	1.39	1.48	1.71	1.60	1.36	1.50	1.56	1.68	1.99
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11510	987	893	1000	983	993	915	908	901	912	1002	1004	1012
	CDD65	6035	522	473	535	533	528	465	443	436	462	537	554	547
	CDH74	60275	5179	4717	5572	5913	5570	4349	3568	3390	4151	5709	6347	5810
(13)	CDH80	18757	1640	1502	1916	2110	1727	1048	664	615	1080	1945	2451	2059
(14)	Wind	WSAvg	2.9	2.6	2.6	2.8	2.8	3.0	3.2	3.3	3.4	3.3	2.9	2.7
(15) Precipitation	PrecAvg	44.1	7.3	7.3	7.5	2.8	4.1	1.5	1.7	1.1	1.8	2.5	3.1	5.1
	PrecMax	69.7	11.5	11.3	15.9	6.1	11.3	5.6	8.7	3.2	6.1	8.1	12.4	13.3
	PrecMin	14.2	3.3	1.8	3.0	1.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.7
	PrecStd	13.7	2.5	2.8	3.9	1.5	3.7	1.8	2.9	1.0	1.8	2.4	3.4	3.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.6	91.4	91.5	90.9	89.9	88.9	87.6	86.9	89.0	90.9	92.3	93.3
		MCWB	78.3	78.8	79.1	78.7	79.0	77.8	77.5	76.4	77.4	77.3	78.8	79.3
	2%	DB	90.1	89.9	90.2	89.8	88.8	87.3	85.8	85.3	87.3	89.7	91.1	91.2
		MCWB	78.6	78.8	78.9	78.8	78.7	77.8	76.9	75.8	76.8	77.7	78.8	79.3
	5%	DB	88.7	88.6	89.1	88.9	87.8	86.2	84.5	84.3	86.2	88.7	90.1	89.8
		MCWB	78.6	78.7	78.7	78.8	78.4	77.5	76.1	75.5	76.5	77.6	78.8	79.2
	10%	DB	87.2	87.3	87.9	88.0	86.8	85.0	83.5	83.3	85.2	87.6	89.2	88.3
		MCWB	78.3	78.4	78.5	78.8	78.1	77.0	75.6	75.2	76.2	77.5	78.7	78.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.8	81.1	80.9	81.0	80.9	80.0	79.3	78.7	79.4	80.5	81.1	81.5
		MCDB	88.4	88.4	88.3	88.2	86.8	85.8	84.8	84.2	85.9	87.3	88.7	89.4
	2%	WB	79.9	80.0	80.1	80.1	80.0	79.1	78.2	77.5	78.5	79.7	80.3	80.5
		MCDB	87.2	87.2	87.4	87.1	86.2	84.9	83.7	83.0	84.7	86.4	88.1	88.6
	5%	WB	79.2	79.3	79.5	79.8	79.5	78.5	77.4	76.8	77.7	79.0	79.9	79.8
		MCDB	86.2	86.1	86.7	86.7	85.6	84.1	82.5	82.2	84.0	85.8	87.5	87.5
	10%	WB	78.7	78.7	79.0	79.2	79.0	77.9	76.7	76.1	77.0	78.3	79.3	79.2
		MCDB	85.4	85.4	86.0	85.9	84.9	83.2	81.6	81.4	83.3	85.0	86.8	86.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.6	11.9	11.3	10.2	9.6	9.3	9.5	10.7	11.7	12.1	11.9
		MCDBR	13.3	13.2	13.2	12.1	11.0	10.7	10.0	10.3	11.4	12.5	13.1	13.6
		MCWBR	5.9	5.8	6.0	5.3	4.9	4.9	4.6	4.7	5.0	5.1	5.4	5.8
	5% WB	MCDBR	12.0	11.9	12.2	11.6	10.4	9.6	9.1	9.2	10.3	11.3	12.2	12.6
		MCWBR	5.8	5.7	6.0	5.5	4.9	4.6	4.5	4.6	4.7	5.1	5.5	5.9
(40) Clear-Sky Solar Irradiance	taub		0.521	0.526	0.492	0.472	0.454	0.441	0.430	0.443	0.459	0.507	0.532	0.527
	taud		2.134	2.126	2.226	2.284	2.331	2.372	2.397	2.330	2.279	2.140	2.084	2.110
	Ebn at Noon		264	262	266	263	258	257	263	267	271	264	259	261
	Edh at Noon		52	53	47	43	39	36	36	40	44	51	55	53
(44) All-Sky Solar Radiation	RadAvg		1665	1725	1757	1730	1630	1549	1618	1790	1985	2035	1860	1629
	RadStd		149	121	135	124	110	87	71	55	85	146	158	147

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	+1.15	N/A	-0.32	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page